

RANISH GHIMIRE

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PROFESSIONAL SUMMARY

Computer Engineering graduate with deep hands-on experience in building and deploying end-to-end AI/ML systems, including real-time CNNs, LSTM-based temporal models, transformer fine-tuning, RAG pipelines, and NLP systems. Experienced in integrating machine learning models into production-ready full-stack applications using FastAPI and React. Additional background in cybersecurity such as SIEM analysis, SOC operations, and threat monitoring, enabling a unique perspective on AI-driven security solutions. Consistent academic topper with seven Gold Medals and ranked Top 1% globally on TryHackMe.

SKILLS

Technical Skills

- **Machine Learning & AI:**

CNN, LSTM, MobileNetV3, EfficientNet, Spectrograms, DSP, RAG, NLP

- **Programming:**

Python, JavaScript, C, C++, SQL, Bash, PowerShell

- **Web Development:**

React, Next.js, Express.js, FastAPI, MongoDB, REST APIs, Supabase, Firebase

- **Cybersecurity:**

SOC Operations, Threat Analysis, Incident Response, SIEM (IBM QRadar, Splunk, ELK, Wazuh), Log Analysis, Vulnerability Assessment

- **Security Tools:**

Nmap, Metasploit, OpenVAS, Nessus, Burp Suite SQLmap, Hydra, Aircrack-ng suite, Netcat, Dradis, Cisco Packet Tracer, Ghidra, IDA Pro, x64dbg

PROJECTS

Voice2Care: Mental Health Monitoring and Support System DEMO Mar 2026

AI/NLP project, Real-time speech emotion recognition and mental health risk detection

- Built a real-time speech emotion recognition and transcription pipeline for live audio analysis.
- Fine-tuned the all-MiniLM-L6-v2 model for improved semantic understanding.
- Integrating custom Emotional RAG using Query Routing operating on 3 different modes.
- Developing a risk detection module to identify violence and self-harm indicators.
- Designing an automated alert system for emergency escalation and rapid response.

POSEPERFECT DEMO RESEARCH PAPER Jan 2025

CNN + LSTM model for real-time exercise detection, rep counting and posture correction

- Developed a real-time fitness tracking system using fine-tuned MobileNetV3Small (CNN) with LSTM (128 units) for temporal sequence modeling.
- Trained on custom self-built dataset of 12090 frames for each exercise samples
- Achieved 92% validation accuracy for exercise detection.
- Implemented repetition counting and posture correction logic using OpenCV and MediaPipe for real-time feedback.
- Integrated the trained model into a React frontend with a FastAPI backend for real-time inference and tracking.

Deep learning model for speech emotion classification across 8 emotion classes

- Extracted log-Mel spectrogram features using digital signal processing for audio emotion classification.
- Fine-tuned MobileNetV3Small and EfficientNet-B3 to compare model performance.
- EfficientNet-B3 achieved 87% validation accuracy, outperforming baseline models.
- Trained and evaluated the model on RAVDESS and TESS emotional speech datasets covering 8 emotion classes.

Scalable MERN stack web application with authentication and cloud integrations

- Built a scalable job portal using the MERN stack, enabling job posting, search, and user role management.
- Designed secure authentication and role-based access control (RBAC) using JWT and HTTP-only cookies.
- Integrated Cloudinary for optimized cloud storage and media management.
- MongoDB Atlas for Backend

EXPERIENCE

- Fine-tuned TTS models such as Kokoro-82M & Omni-voice for Nepali-English, Maithili code-switching under the Speech team.
- Benchmarked 8+ TTS models for latency, throughput, and multilingual quality.
- Debugged cross-platform phonemizer issues in the Kokoro-ONNX inference stack.
- Developed a real-time voice-to-voice pipeline with sub-second latency using LiveKit.
- Extended the voice-to-voice pipeline to support Nepali/Maithili speech interaction with RAG-powered LLM.
- Built and deployed an offline-first FastAPI, PostgreSQL, and Flutter platform for OTP-authenticated Maithili speech data collection, featuring milestone incentives and an admin review dashboard.
- Involved in building a cervical cancer detection system using microscope images and CNN backbones like ConvNeXt.
- Involved in building an AI agent that screens CVs against job descriptions to shortlist candidates.

- Monitored and analyzed security events in IBM QRadar SIEM to identify anomalies and potential incidents.
- Investigated suspicious logs and correlated events to detect threats and minimize false positives.
- Generated 3+ monthly security reports highlighting incident trends and attack patterns.
- Managed user roles, access controls and system configurations in QRadar Admin Console.
- Onboarded log sources and optimized DSM configurations in IBM QRadar to enhance visibility.

EDUCATION

Kantipur Engineering College
Aggregate: 87.60%
🏆 Ranked 1st among all IOE BCT graduates (Batch of 2026)

DAV Sushil Kedia Vishwa Bharati School
GPA - 3.85 / 4

Adarsha Vidya Mandir (A.V.M.)
SEE GPA - 3.8 / 4

ADDITIONAL INFORMATION

Certifications

- Machine Learning, Deep learning, AI Certifications - Coursera
- MERN Stack Certification - Udemy
- Advanced Cybersecurity - Broadwayinfosys
- IBM QRadar SIEM Foundations - IBM
- Certified Ethical Hacker - Cisco
- Networking Basics - Cisco

Awards/Activities

- Received Best Minor Project Award for Poseperfect at Kantipur Engineering College.
 - Won 2nd Runner-Up at Hult Prize 2024.
 - Ranked in the Top 1% globally on TryHackMe
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